NLI-Based Knowledge Coverage as a Swarm Evaluation Metric

Abstract

We present the evaluation metric used to quantify knowledge acquisition in a multi-agent swarm simulation. Each agent maintains a continuously updated natural-language summary of its observations; we measure how much of a fixed ground-truth claim set is semantically covered by that summary using a Natural Language Inference (NLI) cross-encoder model. The resulting scalar score, the *coverage score*, lies in $[0, 1]$ and is tracked for every agent at regular temporal snapshots, enabling both per-agent trajectories and swarm-wide aggregate curves.

1 Experiment Overview

The system places a swarm of N autonomous agents in a bounded 2-D environment seeded with *subject* agents, each carrying a short textual snippet describing a real-world scene (a university career fair). Agents perform a continuous random walk; upon entering the proximity radius of a subject they absorb its snippet, and upon meeting a peer they exchange their current summaries. An LLM fuses the accumulated observations into a single evolving natural-language summary for each agent.

The central research question is whether—and how quickly—a swarm collectively acquires a complete, accurate representation of the scene, and how different communication strategies (self-learning only vs. social learning) affect that process. Answering this requires a metric that can compare an agent's free-form summary against a structured set of ground-truth facts in a semantically meaningful way, motivating the NLI-based coverage score described below.

2 Natural Language Inference and the DeBERTa-v3 Cross-Encoder

Natural Language Inference (NLI) is the task of determining, given a *premise* p and a *hypothesis* h, whether p entails h, contradicts h, or is neutral with respect to h [1]. We use the `cross-encoder/nli-deberta-v3-large` model [2], which concatenates the premise and hypothesis into a single input sequence and passes it through a DeBERTa-v3-large encoder. A linear classification head followed by a softmax layer produces a three-class probability distribution over {contradiction, entailment, neutral}.

The model is used via the Sentence-Transformers `CrossEncoder` API [3]. Given a batch of (p, h) pairs, `CrossEncoder.predict` returns an array of softmax probabilities with a fixed index convention:

$$\mathbf{p}(p, h) = \left[p_{\text{contra}},\ p_{\text{ent}},\ p_{\text{neut}}\right] \ \in\ \mathbb{R}^3, \\ p_{\text{contra}} + p_{\text{ent}} + p_{\text{neut}} = 1. \tag{1}$$

3 Coverage Score: Formal Definition

3.1 Inputs

Let σ denote the current natural-language summary produced by an agent, and let $\mathcal{C} = \{c_1, c_2, \ldots, c_n\}$ be the ground-truth claim set for the current experiment scenario. The claims are fixed short sentences drawn from a curated library (e.g. *"Event staff scan registrations and hand out lanyards at the entrance."*) that collectively describe the scene the agents are exploring.

3.2 Sentence Tokenization

The agent summary σ is split into individual sentences using NLTK's Punkt sentence tokenizer, yielding

$$\mathcal{S}(\sigma) = \{s_1, s_2, \ldots, s_m\}, \tag{2}$$

where $m = |\mathcal{S}(\sigma)|$ varies with summary length.

3.3 Pair Matrix Construction

To check whether the summary supports any given claim, every summary sentence is paired with every claim. The full set of evaluation pairs is

$$\mathcal{P}(\sigma, \mathcal{C})\ =\ \big\{\,(s_i,\ c_j)\ \big|\ s_i \in \mathcal{S}(\sigma),\ c_j \in \mathcal{C}\,\big\}. \tag{3}$$

The matrix has $|\mathcal{P}| = m \times n$ entries, enumerated in *claim-major* order: for each claim c_j all m sentence counterparts are listed consecutively. The complete batch $\mathcal{P}$ is passed to the NLI model in a single forward pass.

3.4 NLI Score Matrix

Applying the model to $\mathcal{P}$ yields a probability matrix

$$\mathbf{P}\ \in\ \mathbb{R}^{n\times m\times 3}, \qquad \mathbf{P}_{j,i} = \mathbf{p}(s_i,\ c_j), \tag{4}$$

where each entry $\mathbf{P}_{j,i}$ is the three-class softmax vector defined in Eq. (1), with s_i as premise and c_j as hypothesis.

3.5 Entailment Decision Per Claim

For each claim c_j we ask: does at least one sentence in the summary constitute a premise that unambiguously entails c_j? This is answered by a hard *argmax* decision across the three output classes for every sentence–claim pair:

$$\hat{\ell}_{j,i} \;=\; \underset{k\,\in\,\{0,1,2\}}{\arg\max}\; \mathbf{P}^{(k)}_{j,i}, \tag{5}$$

where $k = 1$ corresponds to the entailment class. A claim c_j is considered *covered* if the entailment label wins for at least one sentence:

$$\text{covered}(c_j, \sigma) \;=\; \bigvee_{i=1}^{m} \Big[\hat{\ell}_{j,i} = 1\Big]. \tag{6}$$

Using the argmax hard decision avoids introducing a free probability threshold and is consistent with standard NLI evaluation practice [1].

3.6 Coverage Score

The final coverage score is the fraction of ground-truth claims that are covered by the summary:

$$\boxed{\text{Cov}(\sigma, \mathcal{C}) \;=\; \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}[\text{covered}(c_j, \sigma)] \;\in\; [0, 1].} \tag{7}$$

A score of 0 means the summary entails none of the ground-truth claims; a score of 1 means every claim is entailed by at least one sentence of the summary.

4 Application in the Swarm System

4.1 Agent Behaviour and Summary Formation

Each `knowledgeAgent` performs a random walk across a 2-D continuous environment. Two interaction modes drive knowledge accumulation:

- **Subject encounter.** When the agent enters the proximity radius of an information site (a *subject* agent), the site's textual snippet is appended to the agent's private context buffer $\mathbf{p}$ (a bounded deque of capacity K_p).
- **Peer encounter.** When two knowledge agents meet, each appends the other's latest summary to its received-context buffer $\mathbf{t}_{\text{recv}}$ (capacity K_r).

After either interaction, an LLM prompted as a *data-fusion module* merges the agent's prior summary τ_{prev}, its private context, and any received summaries into an updated summary:

$$\tau \;\leftarrow\; \text{LLM}(\tau_{\text{prev}},\; \mathbf{p},\; \mathbf{t}_{\text{recv}})\,. \tag{8}$$

The output is capped at 150 words per prompt instruction.

4.2 Snapshot Evaluation

The simulation environment records a snapshot every Δt seconds. At snapshot time $t \in \{t_1, t_2, \ldots, t_T\}$, each agent a's current summary $\sigma_a(t)$ is evaluated:

$$\text{score}_a(t) \;=\; \text{Cov}(\sigma_a(t),\, \mathcal{C})\,. \tag{9}$$

5 Illustrative Result

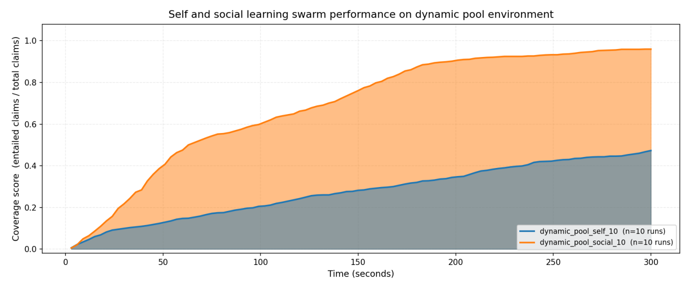

Figure 1: Average coverage score over time: self-learning vs. social learning, averaged across R=10 runs (dynamic-pool environment).

Figure 1 plots the swarm-mean coverage $\bar{S}(t)$ for two experimental configurations in the dynamic-pool environment. Social learning consistently outpaces self-learning, reaching near-complete ground-truth coverage within 200 seconds while the self-learning swarm plateaus below 0.5, demonstrating the metric's sensitivity to inter-agent communication.

References

[1] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning, "A large annotated corpus for learning natural language inference," in *Proc. EMNLP*, 2015, pp. 632–642.

[2] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," *arXiv:2111.09543*, 2021.

[3] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP*, 2019, pp. 3982–3992.